VTPEH 6270 - Check Point 06

Statistical Analyses

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1 Introduction

GitHub repository: https://github.com/xrc577/VTPEH6270-project

1.1 Research Question

Is monthly RNG production associated with digested organic waste inputs levels (low, medium, high)?

1.2 Scientific Plausibility and Context.

Anaerobic digestion converts biodegradable organic matter in wastewater sludge and co-digested substrates into renewable natural gas through a series of microbial processes. The research shows that methane production is strongly influenced by the quantity and characteristics of the organic feedstock supplied to the digester, as higher concentrations of degradable volatile solids increase the potential for biogas generation

(Appels et al. 2008). Co-digestion of wastewater sludge with additional organic substrates, such as food waste, has been widely shown to enhance methane production by increasing organic loading and improving substrate balance (Mata-Alvarez et al. 2014). Evidence also indicates that increasing organic loading can lead to higher methane yields, although the extent of improvement depends on operational stability and process constraints (Li et al. 2024).

Therefore, examining whether renewable natural gas (RNG) production varies across different levels of digested organic waste inputs is mechanistically justified, as variations in organic loading are expected to influence energy recovery performance.

2 Material & Methods

```r
library(dplyr)
library(janitor)
library(ggplot2)
```

2.1 Data

The dataset was obtained from NYC Open Data and includes monthly performance indicators for wastewater treatment facilities operating co-digestion and biogas-to-grid programs.

Data were imported and cleaned in R using the `janitor` and `dplyr` packages. Column names were standardized and key variables were renamed for consistency. Relevant variables include sludge digested, food scraps digested, and renewable natural gas (RNG) production.

A derived variable for food scraps input level was created by categorizing food scraps digested into three groups (low, medium, high) based on tertiles. This allows comparison of RNG production across input levels.

```r
# Import original dataset
data_wastewater <- read.csv(
  "Wastewater Co-digestion and Biogas-to-grid Performance Indicators (New York City).csv"
)

# Clean column names (lowercase + underscore format)
data_wastewater <- clean_names(data_wastewater)

# Rename variables to clean analysis-friendly names
data_wastewater <- data_wastewater %>%
  rename(
    sludge_digested = sludge_digested_wet_tons,
    food_scraps_digested = food_scraps_digested_wet_tons,
    rng_produced = rng_production_mm_btu,
    rng_system_uptime = rng_system_uptime,
    biogas_flared = flared_biogas_mscf,
    flaring_reduction = reduction_in_flaring
  )

# View structure
str(data_wastewater)

# Save the new dataset table to file
write.csv(
```

```
  data_wastewater,
  "wastewater.csv",
  row.names = FALSE
)
```

2.2 Statistical Analyse

To examine whether monthly RNG production differs across food scraps input levels, a boxplot was first used for visualization, followed by a one-way analysis of variance (ANOVA).

The independent variable is food scraps input level (low, medium, high), which was created by categorizing food scraps digested into tertiles. The dependent variable is monthly RNG production.

The null hypothesis (H0) states that mean RNG production is equal across all groups, while the alternative hypothesis (H1) states that at least one group differs.

Model assumptions, including normality of residuals and homogeneity of variance, were evaluated using diagnostic plots.

All analyses were conducted in R.

2.2.1 Create Input Level Variable

To compare RNG production across different levels of organic input, food scraps digested was categorized into three groups (low, medium, high) based on tertiles.

```
data_wastewater <- data_wastewater %>%
  mutate(
    food_scraps_input_level = cut(
      food_scraps_digested,
      breaks = quantile(food_scraps_digested, probs = c(0, 1/3, 2/3, 1), na.rm = TRUE),
      labels = c("low","medium","high"),
      include.lowest = TRUE
    )
  )
```

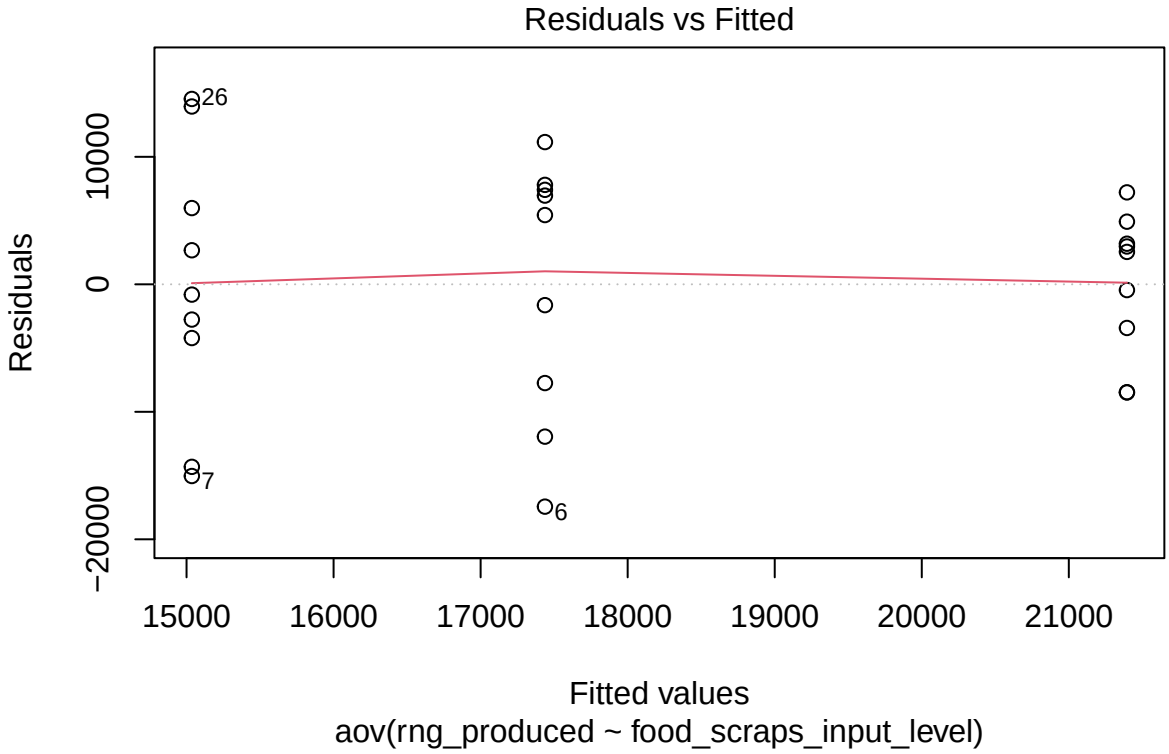
2.2.2 One-way ANOVA

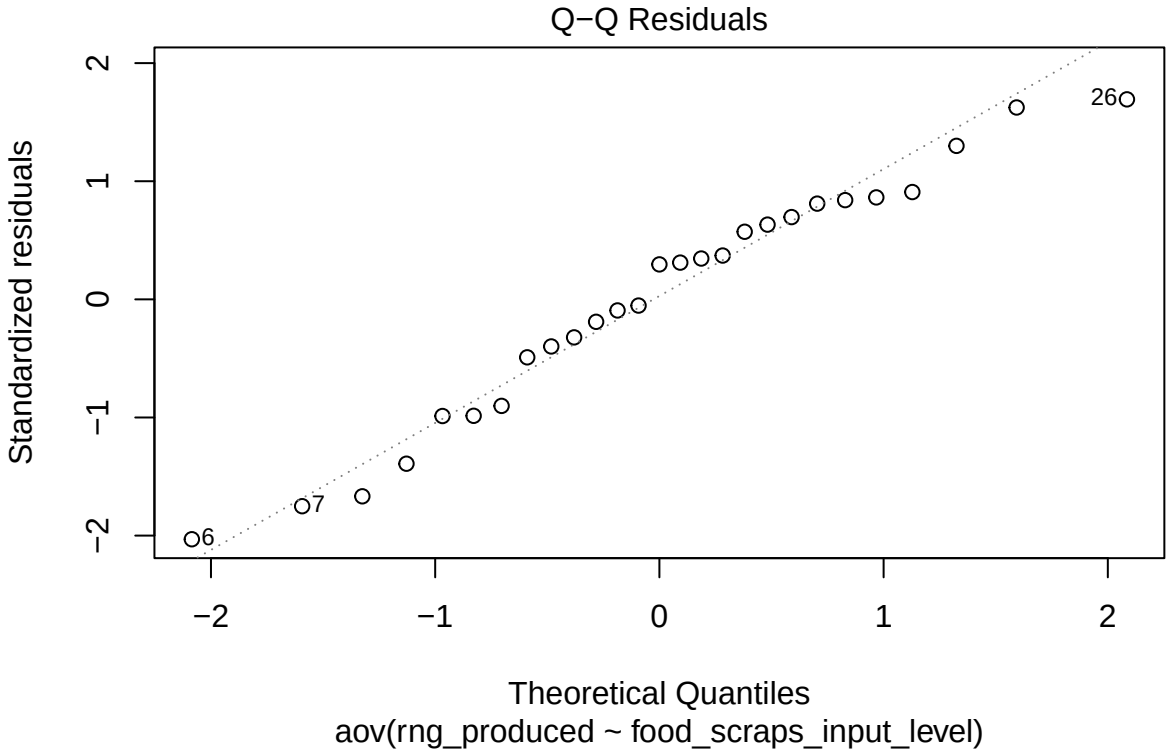
A one-way ANOVA was performed to test whether mean RNG production differs across food scraps input levels.

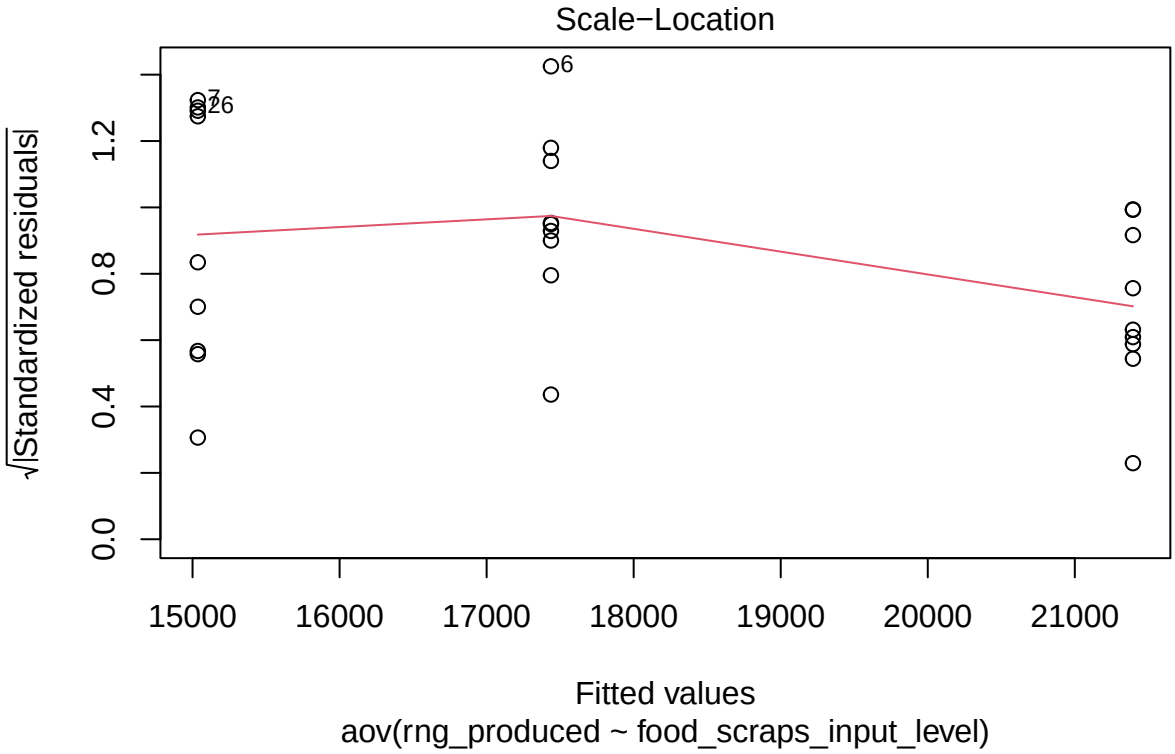
```
anova_model <- aov(rng_produced ~ food_scraps_input_level, data = data_wastewater)
summary(anova_model)
```

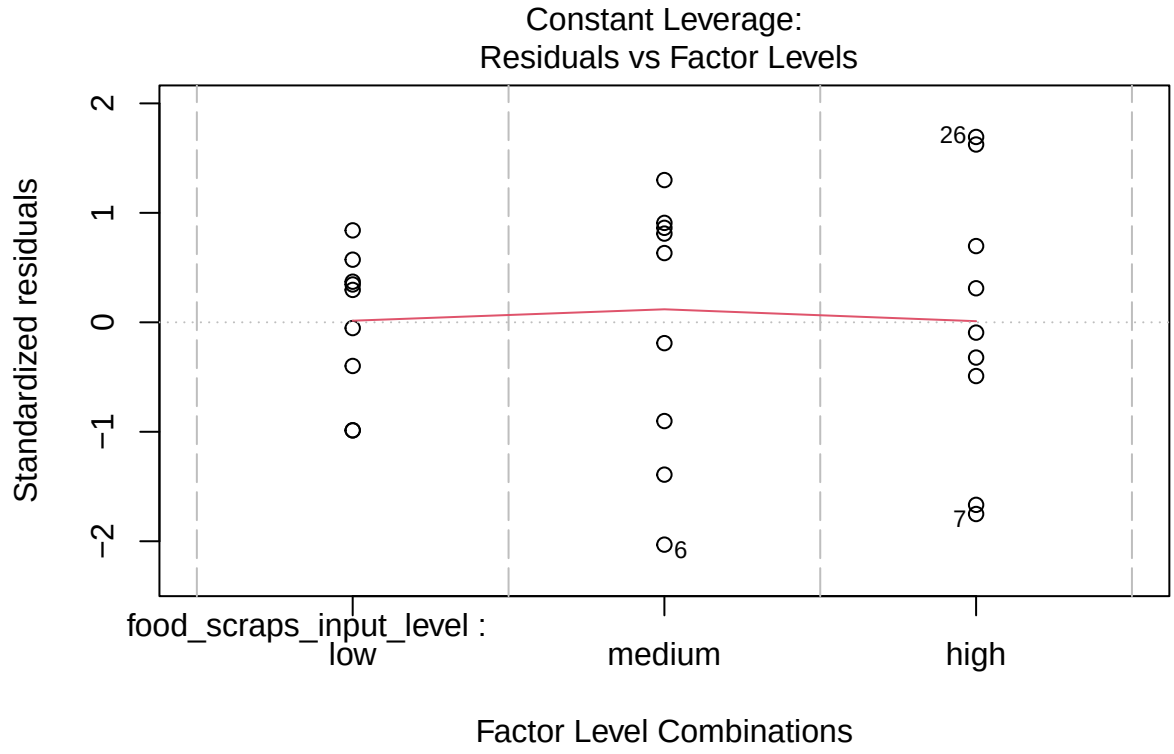
```
##                          Df    Sum Sq  Mean Sq F value Pr(>F)
## food_scraps_input_level   2 1.856e+08 92807009   1.119  0.343
## Residuals                24 1.991e+09 82941850
```

```
plot(anova_model)
```


Residuals vs Fitted
Residuals
10000
0
−20000
26
7
6
15000
16000
17000
18000
19000
20000
21000
Fitted values
aov(rng_produced ~ food_scraps_input_level)


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
aov(rng_produced ~ food_scraps_input_level)
26
7
6
−2
−1
0
1
2

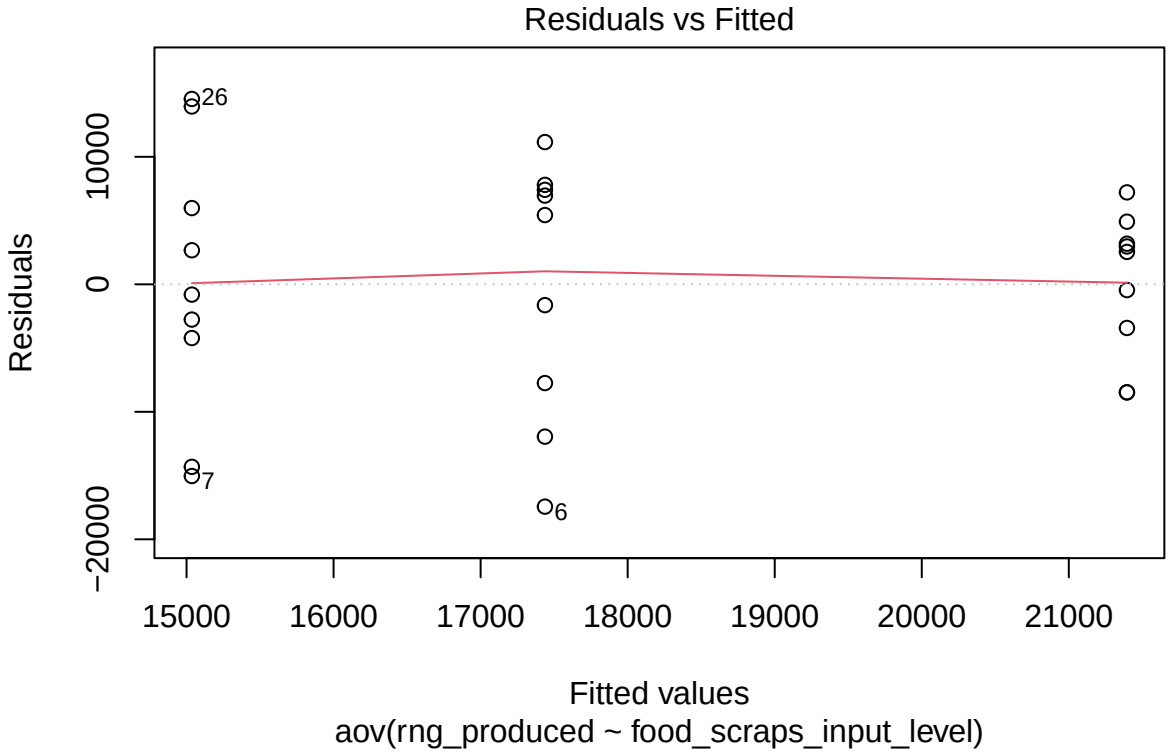

Scale−Location
√|Standardized residuals|
0.0
0.4
0.8
1.2
6
7
26
15000
16000
17000
18000
19000
20000
21000
Fitted values
aov(rng_produced ~ food_scraps_input_level)

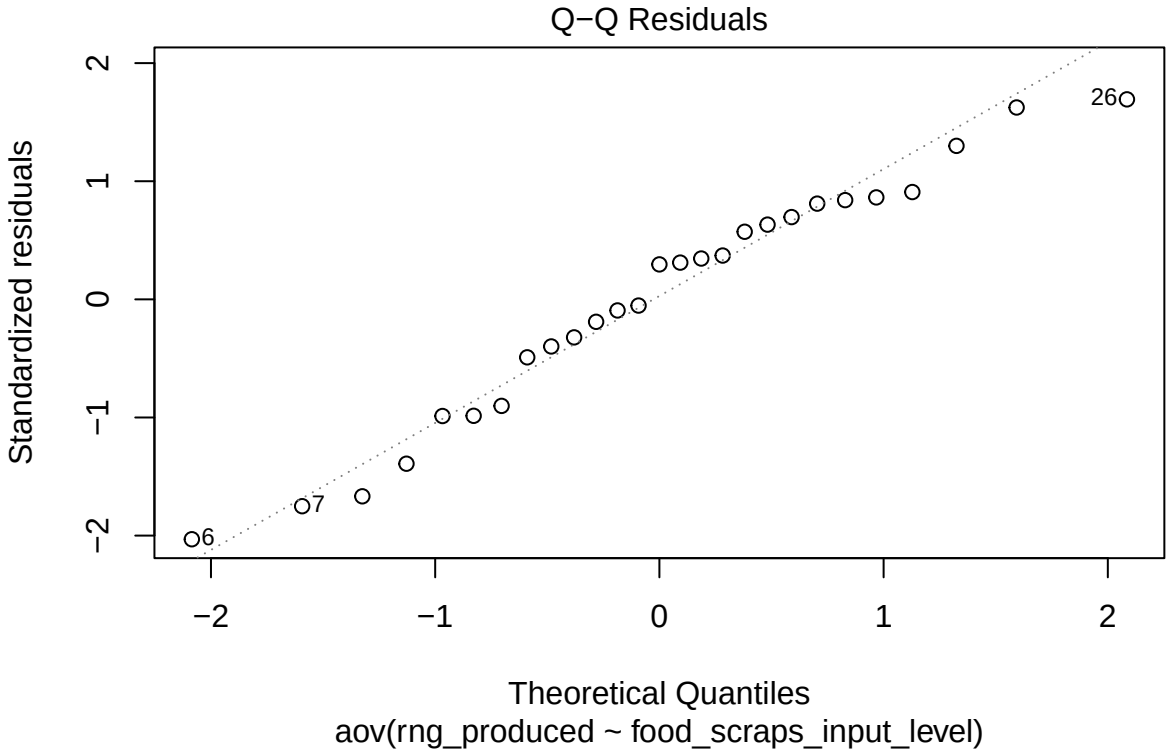


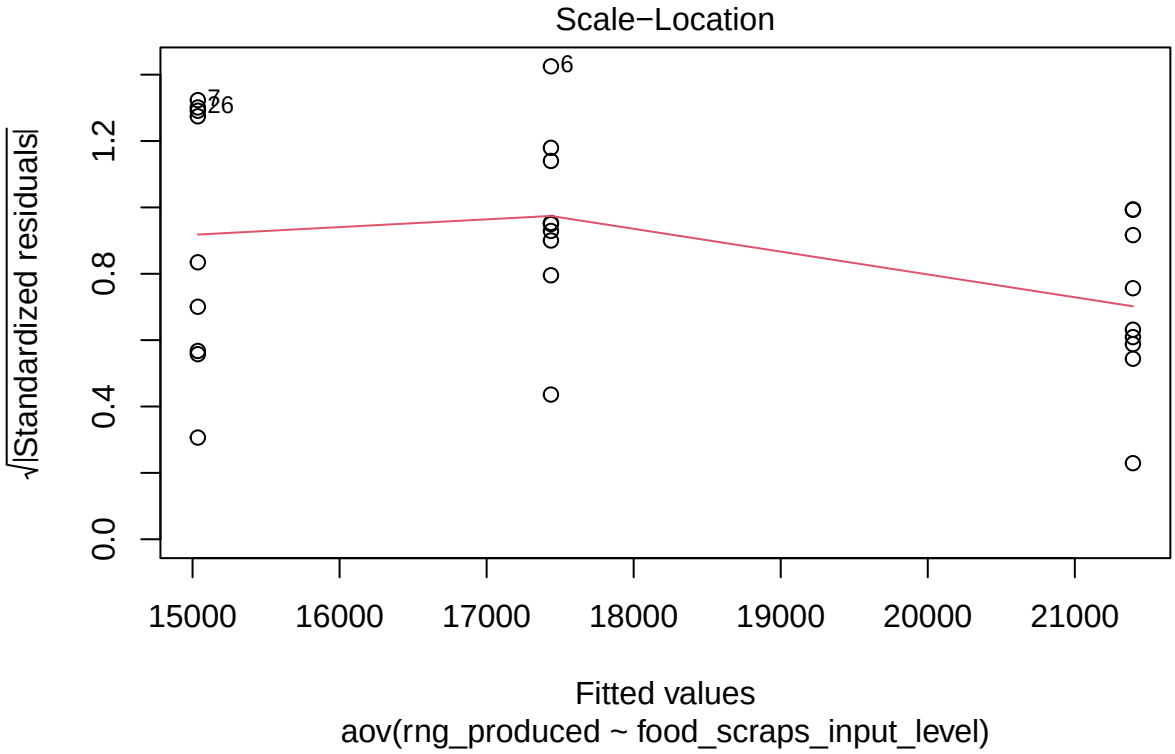
2.2.3 Diagnostic Plots

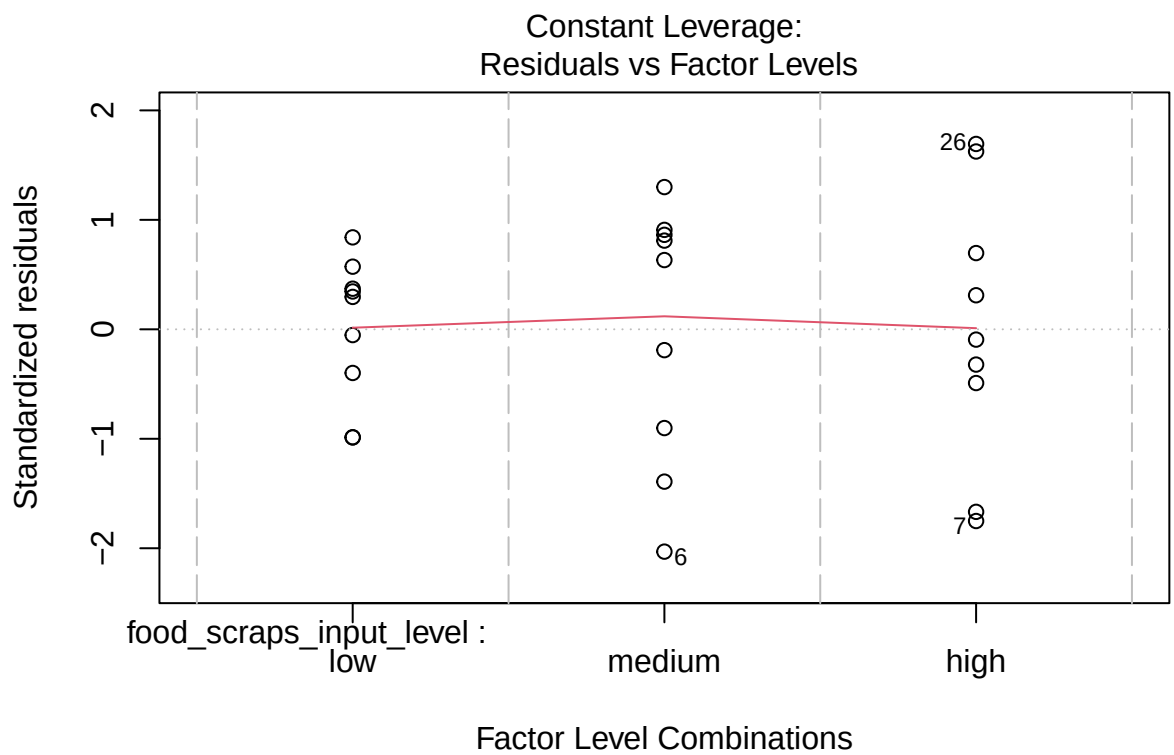
Diagnostic plots were used to evaluate model assumptions, including normality of residuals and homogeneity of variance.

```
plot(anova_model)
```


Residuals vs Fitted
Residuals
10000
0
−20000
26
7
6
15000
16000
17000
18000
19000
20000
21000
Fitted values
aov(rng_produced ~ food_scraps_input_level)


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
aov(rng_produced ~ food_scraps_input_level)
26
7
6
−2
−1
0
1
2

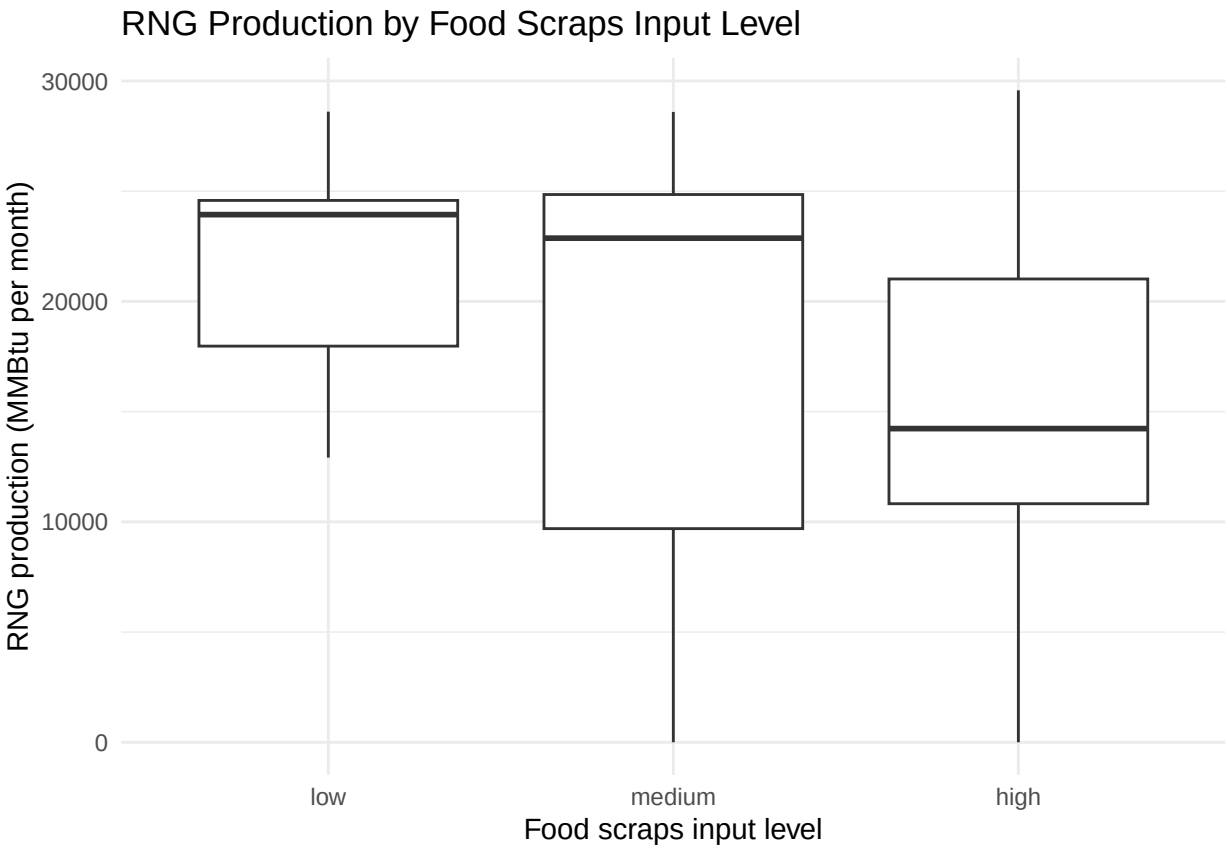

Scale−Location
√|Standardized residuals|
0.0
0.4
0.8
1.2
6
7
26
15000
16000
17000
18000
19000
20000
21000
Fitted values
aov(rng_produced ~ food_scraps_input_level)



2.2.4 Visualization of RNG Production

A boxplot was created to visualize the distribution of RNG production across input levels.

```
ggplot(data_wastewater, aes(x = food_scraps_input_level, y = rng_produced)) +
  geom_boxplot() +
  labs(
    title = "RNG Production by Food Scraps Input Level",
    x = "Food scraps input level",
    y = "RNG production (MMBtu per month)"
  ) +
  theme_minimal(
  )
```


RNG Production by Food Scraps Input Level
30000
20000
10000
0
RNG production (MMBtu per month)
low
medium
high
Food scraps input level

3 Results

3.1 Visualization

3.1.1 RNG Production by Food Scraps Input Level Boxplot

3.2 Statistical Analyses

A boxplot was used as a visualization to examine the distribution of monthly RNG production across food scraps input levels (low, medium, high). This approach allows for direct comparison of central tendency and variability between groups.

To formally test whether mean RNG production differs across input levels, a one-way analysis of variance (ANOVA) was performed. ANOVA was selected as the independent variable is categorical with three groups, and the outcome variable (RNG production) is continuous, making it appropriate for comparing group means.

The ANOVA results indicated that there was no statistically significant difference in mean RNG production across the three groups ($F = 1.12$, $p = 0.343$).

Diagnostic plots were used to evaluate model assumptions, including normality of residuals and homogeneity of variance. Visual inspection suggested no major violations of these assumptions.

Although the boxplot shows some variation in RNG production across input levels, substantial variability is observed within groups.

4 Conclusions

This analysis aimed to evaluate whether food scraps input level influences monthly RNG production. The findings suggest that variation in input level alone may not be a strong determinant of RNG output.

One possible explanation is that RNG production is influenced by multiple operational and environmental factors, such as digestion efficiency, system uptime, and feedstock composition, which may vary within each input group. This variability could mask potential differences associated with input levels.

These results indicate that simply increasing food scraps input may not directly lead to higher RNG production, and highlight the importance of considering additional system-level factors in optimizing biogas generation.

5 AI Use Disclosure Statement

This document was generated with the assistance of AI-based tools, including ChatGPT, which were used to drafting code and identifying relevant functions. All final decisions, analyses, and revisions were independently reviewed and completed by the author.

Appels, Lise, Jan Baeyens, Jan Degrève, and Raf Dewil. 2008. "Principles and Potential of the Anaerobic Digestion of Waste-Activated Sludge." *Progress in Energy and Combustion Science* 34 (December): 755–81. https://doi.org/10.1016/j.pecs.2008.06.002.

Li, Pengfei, Hao Zhao, Chongbo Cheng, Tingting Hou, Dekui Shen, and Youzhou Jiao. 2024. "A Review on Anaerobic Co-Digestion of Sewage Sludge with Other Organic Wastes for Methane Production: Mechanism, Process, Improvement and Industrial Application." *Biomass and Bioenergy* 185 (May): 107241. https://doi.org/10.1016/j.biombioe.2024.107241.

Mata-Alvarez, J., J. Dosta, M. S. Romero-Güiza, X. Fonoll, M. Peces, and S. Astals. 2014. "A Critical Review on Anaerobic Co-Digestion Achievements Between 2010 and 2013." *Renewable and Sustainable Energy Reviews* 36 (August): 412–27. https://doi.org/10.1016/j.rser.2014.04.039.